

Automatic Rigidity for Collatz-Invariant 5-Automatic Sequences

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Lean 4 formally verified version (the main ARC5 proof does not use Cobham's theorem)

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Abstract

Let $a = (a_n)_{n \geq 0}$ be a sequence over a finite alphabet. We prove that if a is 5-automatic and satisfies

$$a_{2n} = a_n, \quad a_{2n+1} = a_{3n+2} \quad (n \geq 0),$$

then a is constant on the positive integers, while a_0 remains free. Equivalently, every finite-alphabet 5-automatic coloring invariant under the shortcut Collatz map is trivial on $\mathbb{N}_{>0}$.

The base-5 case requires a mechanism different from the base-2 and base-3 ARC results. Direct transfer of the 5-kernel to a multiplicatively independent kernel does not close under the odd Collatz branch. Instead, shortcut invariance first yields a strong local-density/coalescence principle: inside every prescribed dyadic residue class one can find a deeper dyadic class on which large finite blocks are synchronized. This forces disagreement sets, and in particular the adjacent-change set $\{n : a_n \neq a_{n+1}\}$, to be dyadically nowhere thick. On the automatic side, a finite-state pumping argument shows that an infinite 5-automatic Boolean support cannot be dyadically nowhere thick. Hence the adjacent-change set is finite, so a is eventually constant. Finally, shortcut invariance under doubling propagates the tail value back to every positive integer.

The complete proof chain was formalized in Lean 4 / Mathlib, culminating in the theorem `arc5_final`. The final entry point assumes only shortcut invariance and finiteness of the base-5 kernel. No instance of Cobham's theorem is used in the ARC5 main proof. As in the earlier ARC papers, the Collatz-basin corollary obtained from the rigidity theorem is a reformulation of the Collatz conjecture, not a proof of it.

Keywords: Collatz conjecture; automatic sequence; finite automata; 5-kernel; invariant coloring; pumping; 2-adic topology; local density; Lean 4.

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Author's note: how this study arose

This work is the third stage of an exploration begun with automatic rigidity in bases 2 and 3. The base-3 proof suggested testing base 5 next, but the first natural attempt failed for a structural reason: if

$$u_n = a_{5^j n + r},$$

then binary decimation produces indices with coefficient $2 \cdot 5^j$, and the odd Collatz identity changes this to a coefficient $3 \cdot 5^j$, which is not a power of 5. Thus the direct kernel-closure mechanism that makes ARC3 short does not survive.

Rather than treating that obstruction as merely technical, the project was reorganized around a different question: what local synchronization is forced by Collatz invariance itself, before automaticity is used? The resulting Lean development grew through a sequence of local-density, dyadic-cylinder, finite-state, pumping, and change-set stages. The decisive bridge is that Collatz invariance makes the change set 2-adically thin, whereas an infinite base-5 automatic support cannot remain that thin.

I am not a professionally trained mathematician. The project has been developed through sustained discussion with a contemporary conversational AI, with repeated adversarial checking and Lean compilation used to expose and repair gaps. The human-readable proof below is intentionally compressed relative to the formal development; the Lean files expand the delicate parity-word, cylinder, pumping, and interface arguments into machine-checked steps.

1 Introduction

Define the shortcut Collatz map $T : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ by

$$T(2n) = n, \quad T(2n+1) = 3n+2 \quad (n \geq 0). \quad (1)$$

On positive integers this is the usual $3x+1$ map with one division by 2 incorporated into the odd step. For background on the $3x+1$ problem, see Lagarias [10].

Let Δ be a finite alphabet. A coloring $a : \mathbb{N}_0 \rightarrow \Delta$ is *T-invariant* if

$$a_n = a_{T(n)} \quad (n \geq 0).$$

By (1), this is equivalent to

$$a_{2n} = a_n, \quad a_{2n+1} = a_{3n+2} \quad (n \geq 0). \quad (2)$$

Finite-state descriptions of Collatz-related sets and relations have been studied from several directions. Shallit and Wilson [11] and Stérin [12] studied automata describing predecessor/ancestor families with bounded odd-step complexity. Stérin and Woods [13] exhibited an explicit base-conversion phenomenon inside the Collatz process. Caucal and Rispal [3] studied transducers for powers of Collatz-type maps, while Dhiman and Pandey [5, 6] established non-definability results for transition and reachability relations in Büchi-arithmetic settings.

The present problem is different. We do not ask an automaton to compute T or to recognize a bounded-depth orbit family. We ask whether a finite-valued coloring that is constant along every edge of the full functional graph of T can itself be 5-automatic without being trivial.

Theorem 1.1 (Automatic rigidity in base 5; ARC5). *Let Δ be a finite alphabet and let $a = (a_n)_{n \geq 0} \in \Delta^{\mathbb{N}_0}$ be a 5-automatic sequence. Assume (2) for every $n \geq 0$. Then there exists $c \in \Delta$ such that*

$$a_n = c \quad (n \geq 1).$$

The value a_0 is arbitrary.

Theorem 1.1 makes no assumption about the truth of the Collatz conjecture. Its proof separates into two forces:

1. *Collatz local synchronization*: invariant colorings contain synchronized blocks inside arbitrarily prescribed dyadic neighborhoods;
2. *odd-base finite-state pumping*: an infinite 5-automatic support cannot avoid an entire dyadic child inside every dyadic neighborhood.

Their combination makes the adjacent-change set finite. Eventual constancy then follows by elementary order theory, and doubling invariance upgrades eventual constancy to constancy on all positive inputs.

2 Automatic sequences, kernels, and the base-5 obstruction

For an integer $k \geq 2$ and a sequence $u = (u_n)_{n \geq 0}$, define the k -kernel

$$K_k(u) := \{(u_{k^e n + r})_{n \geq 0} : e \geq 0, 0 \leq r < k^e\}.$$

A finite-alphabet sequence is k -automatic if and only if its k -kernel is finite [1]. We use the usual most-significant-digit (MSD) convention when passing to DFAO presentations; leading-zero normalization is handled explicitly in the formal bridge.

For ARC3, the identities (2) imply direct closure of the full 3-kernel under binary decimation, hence $K_2(a) \subseteq K_3(a)$. Cobham's theorem then supplies eventual periodicity. No analogous one-line closure holds in base 5. Indeed, for

$$u_n = a_{5^j n + r},$$

the odd binary decimation gives

$$u_{2n+1} = a_{2 \cdot 5^j n + (5^j + r)}.$$

When the index is odd, the odd branch of (2) changes a term a_{2m+1} to a_{3m+2} , introducing the coefficient $3 \cdot 5^j$. This is not a base-5 kernel coefficient. The failure is therefore algebraic, not merely a missing simplification.

The ARC5 proof avoids forcing a cross-base kernel inclusion. Instead it extracts 2-adic local structure from the Collatz map itself and then lets base-5 finite-state pumping collide with that structure.

3 Collatz local coalescence and synchronized blocks

This section states the local-density component in the form used later. The full formalization develops it from parity words and their affine branch maps.

For a parity word $w = (\varepsilon_0, \dots, \varepsilon_{L-1}) \in \{0, 1\}^L$, let F_w denote the affine composition obtained by applying

$$E(x) = \frac{x}{2} \quad \text{for } \varepsilon_i = 0, \quad O(x) = \frac{3x+1}{2} \quad \text{for } \varepsilon_i = 1.$$

If $k(w)$ is the number of odd letters, then

$$F_w(x) = \frac{3^{k(w)}(x + R(w))}{2^L} \tag{3}$$

for a rational normalized shift $R(w)$. The Stage-1/2 part of the formalization constructs, for each $N > 0$, two equal-length words u, v with the same odd-step count and

$$R(u) - R(v) = N.$$

Equation (3) then gives the affine identity

$$F_u(x) = F_v(x + N) \tag{4}$$

identically in x .

The essential arithmetic issue is admissibility: the formal affine branches must agree with the actual parities encountered by the integer shortcut map. A parity word of length L is realized on a residue class modulo 2^L . Refining dyadic cylinders allows the two words in (4) to be realized simultaneously.

Lemma 3.1 (Universal dyadic-cylinder coalescence). *For every integer $N > 0$ there exist a depth L and, inside any prescribed dyadic residue class, a deeper nonempty dyadic residue class C such that*

$$T^L(x) = T^L(x + N) \quad (x \in C).$$

The statement is first proved over $\mathbb{Z}$ and then restricted to positive natural inputs.

Compressed proof. The parity-word construction gives universal affine coalescence (4). Dyadic refinement realizes the two branch words on a common seed cylinder. The formal proof then balances prefixes so that the two endpoint affine maps have equal length and equal odd-step count. If the remaining endpoint gap is D , there are three cases. For $D = 0$ the prefixes already coalesce. For $D > 0$, append a universal coalescing word pair for D ; for $D < 0$, do the symmetric construction with $-D$. Seed coalescence is then promoted to uniform coalescence on the whole final subcylinder because the branch data are constant on that cylinder. The natural-domain bridge proves that the integer shortcut map preserves nonnegativity and that every dyadic cylinder contains a positive natural representative. All of these substeps are formalized in the Stage-1 through Stage-4P files. $\square$

If a is T -invariant, Lemma 3.1 immediately converts orbit coalescence into color equality.

Proposition 3.2 (Strong dyadic synchronization). *Let a satisfy (2). Fix a finite set of offsets $J \subset \mathbb{N}_0$ and a dyadic parent class $n \equiv r \pmod{2^m}$. Then there is a nonempty positive dyadic child class $n \equiv t \pmod{2^M}$ contained in the parent on which*

$$a_n = a_{n+j} \quad (j \in J)$$

holds simultaneously for every n in the child.

Proof idea. Apply Lemma 3.1 successively to the finitely many positive gaps in J , refining the cylinder after each step. Because each refinement stays inside the previous parent, all equalities survive simultaneously on the final child. The Lean development packages the result as finite simultaneous local density and then as a strong dyadic synchronization interface. $\square$

A useful global consequence is the existence of arbitrarily long monochromatic intervals.

Corollary 3.3 (Finite invariant blocks). *For every $q \geq 0$ there is a positive integer n such that*

$$a_n = a_{n+j} \quad (0 \leq j < 5^q).$$

Moreover one can choose a positive k so that the whole aligned block

$$a_{5^q k} = a_{5^q k + r} \quad (0 \leq r < 5^q) \tag{5}$$

has one color.

Proof. The first statement is Proposition 3.2 applied to $J = \{0, 1, \dots, 5^q - 1\}$. For the aligned version, first obtain a monochromatic interval of length 5^{q+1} . The next multiple of 5^q after its left endpoint lies at distance at most 5^q , so the complete following block of length 5^q remains inside the longer monochromatic interval. $\square$

The crucial point is that no automaticity has been used in this section.

4 Dyadic thinness of disagreement sets

At exact base-5 level q , write

$$K_{5,q}(a) := \{v_r : v_r(n) = a_{5^q n + r}, 0 \leq r < 5^q\}.$$

When a is 5-automatic, the full union $\bigcup_q K_{5,q}(a)$ is finite. Digit transitions $v(n) \mapsto v(5n + d)$, $0 \leq d < 5$, remain inside this finite graph.

The local synchronization from Section 3 can be transported through multiplication by 5^q . Since 5^q is coprime to every power of 2, multiplication by 5^q permutes dyadic residue classes at each fixed depth. Consequently, for every exact level q , every pair $v, w \in K_{5,q}(a)$, and every dyadic parent class, there is a deeper nonempty positive child class on which v and w agree pointwise.

This motivates the following definition.

Definition 4.1 (Dyadically nowhere thick). A set $S \subseteq \mathbb{N}_0$ is *dyadically nowhere thick* if, for every $m, r \geq 0$, there exist $M, t \geq 0$ such that

1. every $n \equiv t \pmod{2^M}$ also satisfies $n \equiv r \pmod{2^m}$;
2. the child class $t \pmod{2^M}$ contains a positive natural number;
3. no $n \equiv t \pmod{2^M}$ belongs to S .

Thus every dyadic neighborhood contains a nonempty positive dyadic sub-neighborhood disjoint from S .

For two states v, w , define the disagreement set

$$\mathcal{E}(v, w) := \{n \geq 0 : v(n) \neq w(n)\}.$$

Proposition 4.2 (Thin disagreement). *Under the hypotheses of Theorem 1.1, for every q and every $v, w \in K_{5,q}(a)$, the set $\mathcal{E}(v, w)$ is dyadically nowhere thick.*

Proof. Given a dyadic parent, the cross-base synchronization described above supplies a nonempty child on which $v(n) = w(n)$ for every input in that child. Hence the child is disjoint from $\mathcal{E}(v, w)$, exactly as in Definition 4.1. $\square$

The Boolean disagreement indicator

$$\delta_{v,w}(n) = \begin{cases} 1, & v(n) \neq w(n), \\ 0, & v(n) = w(n), \end{cases}$$

is again 5-automatic. Formally, all its base-5 residuals belong to a finite comparison family built from pairs of states in the finite 5-kernel. This observation will also be used for adjacent changes.

5 Topological pumping for 5-automatic supports

We now isolate the finite-state theorem that replaces Cobham's theorem in ARC5.

Theorem 5.1 (Odd-base topological pumping, base 5). *Let $s : \mathbb{N}_0 \rightarrow \{0, 1\}$ be 5-automatic and let*

$$S = \{n : s(n) = 1\}.$$

If S is dyadically nowhere thick, then S is finite.

The terminology “topological pumping” is used here for the finite-state/p-adic interaction in the following proof; it is not intended as a claim of standard nomenclature.

Proof. Assume that S is infinite and choose a finite MSD base-5 DFAO $\mathcal{M}$ generating s . Because an infinite subset of $\mathbb{N}_0$ is unbounded, S contains a positive integer whose canonical base-5 word x has length at least the number of states of $\mathcal{M}$. By the finite-state pigeonhole argument, write

$$x = abc, \quad b \neq \varepsilon,$$

so that the state after reading a is the same as the state after reading ab . Hence every pumped word

$$ab^k c \quad (k \geq 0)$$

is also accepted into the support.

Let $L = |b|$. The numerical values of the pumped words have the form

$$N_k = N_0 + d G_k, \quad G_k = 1 + 5^L + \dots + 5^{L(k-1)}, \quad (6)$$

with a positive integer increment $d > 0$. Positivity uses that x is a canonical positive word and that the pumped block is nonempty. Write

$$d = 2^s(2v + 1).$$

Because 5^L is odd and in fact $5^L \equiv 1 \pmod{4}$, differences of the geometric sums in (6) have exactly controlled powers of 2. The formal arithmetic pumping lemma yields: for every $k, t, u \geq 0$ there exists $w \geq 0$ such that

$$N_{k+2^t(2u+1)} = N_k + 2^{s+t}(2w + 1). \quad (7)$$

This exact two-factorization is the key point: pumping can move from a point in the support into every sufficiently deep dyadic child of the parent class modulo 2^s .

More concretely, after normalizing by 2^s , the odd increment in (6) induces a permutation of residues modulo 2^R over a block of 2^R pump indices. Restoring the factor 2^s and the offset N_0 shows that the pumped family intersects every child class

$$N_0 + 2^s r \pmod{2^{s+R}}, \quad 0 \leq r < 2^R.$$

Thus one dyadic parent class has no deeper child class disjoint from S . This contradicts dyadic nowhere-thickness. Therefore S must be finite. $\square$

Remark 5.2. The proof uses the oddness of the base in an essential way: powers of 5 are units modulo powers of 2, and the pumped geometric family can therefore spread through dyadic children. The theorem is proved in the Lean development first for an explicit MSD-DFAO and then transferred from finite 5-kernel automaticity to the MSD presentation.

Combining Proposition 4.2 with Theorem 5.1 shows that every same-level disagreement set $\mathcal{E}(v, w)$ is finite. For the final ARC5 theorem, however, an even more direct object is available: the adjacent-change set.

6 The adjacent-change set and eventual constancy

Define

$$\mathcal{D}(a) := \{n \geq 0 : a_n \neq a_{n+1}\}. \quad (8)$$

Let $\chi_{\mathcal{D}}$ be its Boolean characteristic sequence.

Proposition 6.1 (Automaticity of adjacent changes). *If a has finite 5-kernel, then $\chi_{\mathcal{D}}$ also has finite 5-kernel.*

Proof. A base-5 residual of $\chi_{\mathcal{D}}$ compares

$$a_{5^q n+r} \quad \text{with} \quad a_{5^q n+r+1}.$$

The first sequence is a 5-kernel state. The second is either a kernel state at the same level or the one-step successor of such a state across the final residue boundary. Since the kernel is finite, the union of all kernel states and all one-step successors is finite. Pairwise Boolean disagreement indicators from this finite comparison family form a finite set, and every residual of $\chi_{\mathcal{D}}$ belongs to it. Hence the 5-kernel of $\chi_{\mathcal{D}}$ is finite. $\square$

Proposition 6.2 (The change set is dyadically nowhere thick). *If a is T -invariant, then $\mathcal{D}(a)$ is dyadically nowhere thick.*

Proof. Fix a dyadic parent class. Apply the strong synchronization property with offsets 0 and 1 (the formal development obtains this as the $q = 1$ case of a stronger block statement). We obtain a nonempty positive dyadic child on which

$$a_n = a_{n+1}$$

for every n in the child. That child is disjoint from $\mathcal{D}(a)$. $\square$

Corollary 6.3 (Finite change set). *Under the hypotheses of Theorem 1.1, $\mathcal{D}(a)$ is finite.*

Proof. By Proposition 6.1, the Boolean support $\mathcal{D}(a)$ is 5-automatic. By Proposition 6.2, it is dyadically nowhere thick. Apply Theorem 5.1. $\square$

Proposition 6.4 (Eventual constancy). *Under the hypotheses of Theorem 1.1, there exist $N \geq 0$ and $c \in \Delta$ such that*

$$a_n = c \quad (n \geq N).$$

Proof. A finite subset of $\mathbb{N}_0$ is bounded. Choose N larger than every element of $\mathcal{D}(a)$. Then $a_n = a_{n+1}$ for every $n \geq N$. Induction gives $a_n = a_N$ for all $n \geq N$. $\square$

7 From the constant tail to all positive integers

The final step is elementary but important: the Collatz invariance itself transports the tail value backwards without invoking any global Collatz convergence statement.

Lemma 7.1 (Doubling invariance). *For every $k, n \geq 0$,*

$$a_{2^k n} = a_n.$$

Proof. The identity $a_{2m} = a_m$ is the even half of (2). Iterate it k times. $\square$

Proposition 7.2 (Tail-to-positive propagation). *Suppose a is T -invariant and $a_n = c$ for every $n \geq N$. Then $a_n = c$ for every $n \geq 1$.*

Proof. Fix $n \geq 1$. The number

$$x = 2^N n$$

satisfies $x \geq N$ (indeed $N \leq 2^N \leq 2^N n$). Hence $a_x = c$. By Lemma 7.1, $a_x = a_n$, so $a_n = c$. $\square$

Proof of Theorem 1.1. Proposition 6.4 gives a constant tail. Proposition 7.2 makes the sequence constant on all positive integers.

Finally, $T(0) = 0$ and no identity in (2) links index 0 to a positive index. Given arbitrary $c, d \in \Delta$, the sequence

$$a_0 = d, \quad a_n = c \quad (n \geq 1)$$

is 5-automatic and satisfies (2). Thus a_0 is genuinely free. $\square$

8 Consequences for Collatz-invariant 5-automatic sets

A set $A \subseteq \mathbb{N}_0$ is called 5-automatic if its characteristic sequence is 5-automatic.

Corollary 8.1 (Invariant 5-automatic sets are trivial on $\mathbb{N}_{>0}$). *Let $A \subseteq \mathbb{N}_0$ be 5-automatic and suppose*

$$n \in A \iff T(n) \in A \quad (n \geq 0).$$

Then either

$$A \cap \mathbb{N}_{>0} = \emptyset \quad \text{or} \quad A \cap \mathbb{N}_{>0} = \mathbb{N}_{>0}.$$

Membership of 0 is arbitrary.

Proof. Apply Theorem 1.1 to the characteristic sequence of A . $\square$

Corollary 8.2 (Nontrivial invariant sets are not 5-automatic). *Let $A \subseteq \mathbb{N}_{>0}$ satisfy $n \in A \iff T(n) \in A$ for all $n \geq 1$. If A is neither empty nor all of $\mathbb{N}_{>0}$, then its characteristic sequence is not 5-automatic.*

Define the basin of 1 by

$$B := \{n \geq 1 : \exists k \geq 0, T^k(n) = 1\}. \quad (9)$$

We index its characteristic sequence on $\mathbb{N}_0$ and put $0 \notin B$.

Lemma 8.3. *For every $n \geq 0$,*

$$n \in B \iff T(n) \in B.$$

Proof. For $n = 0$ both sides are false. For $n \geq 1$, if $T(n)$ reaches 1, then n reaches 1 one step later. Conversely, if n reaches 1 and $n \neq 1$, delete the first step of the trajectory. For $n = 1$, note that $T(1) = 2$ and $T(2) = 1$. $\square$

Corollary 8.4 (A 5-automaticity criterion for the Collatz conjecture). *The Collatz conjecture on the positive integers holds if and only if the characteristic sequence of B is 5-automatic.*

Proof. If the Collatz conjecture holds, then $B = \mathbb{N}_{>0}$, whose characteristic sequence is automatic in every base. Conversely, if B is 5-automatic, Lemma 8.3 and Corollary 8.1 imply that B is either empty or all of $\mathbb{N}_{>0}$. Since $1 \in B$, we get $B = \mathbb{N}_{>0}$. $\square$

Remark 8.5. Corollary 8.4 is a reformulation of the Collatz conjecture, not a proof. Theorem 1.1 does not show that the basin is 5-automatic. It says that if the Collatz conjecture is false, then the full basin of 1 must have genuinely non-5-automatic global complexity.

9 Relation to previous work and novelty audit

The automata literature around Collatz contains several nearby but distinct objects. Shallit and Wilson [11] and Stérin [12] give finite-state descriptions of predecessor/ancestor families under bounded odd-step restrictions. Stérin and Woods [13] emphasize the interaction between bases 2 and 3 encoded by the Collatz process. Caucal and Rispal [3] construct transducer descriptions of Collatz-type functions and powers. Dhiman and Pandey [5, 6] study logical definability of two-variable transition/reachability relations. These are important context, but they are not classifications of arbitrary finite-valued invariant colorings.

Recognizable and affinely generated sets have also been studied by Klarner [9], Garth and Gouge [7], and Kärki, Lacroix, and Rigo [8]. Their closure-generation problems differ from the two-sided orbit-saturation condition

$$n \in A \iff T(n) \in A.$$

The ARC5 proof also uses a finite-state pumping mechanism. The ordinary pumping phenomenon for regular languages is classical; the point specific to the present argument is its arithmetic interaction with powers of 2 through the numerical values of repeated base-5 blocks. The theorem needed here is not merely that an infinite regular language can be pumped, but that an infinite base-5 automatic support must penetrate all sufficiently deep dyadic children of some dyadic parent. This is what contradicts Definition 4.1.

For the novelty audit, targeted searches through September 17, 2026 were carried out using formulations involving 5-automatic Collatz-invariant sequences, recognizable invariant sets, finite-state Collatz colorings, and automatic disagreement/change sets. We did not locate a prior theorem explicitly asserting Theorem 1.1, nor a prior result stated in the same dyadic-nowhere-thick pumping form used in Theorem 5.1. This is a provisional literature conclusion, not a proof of exhaustive novelty; an equivalent consequence of a more general theorem under different terminology cannot be ruled out completely.

10 Comparison with ARC2 and ARC3

The three ARC cases reach the same positive-domain rigidity conclusion by substantially different bridges.

	ARC2	ARC3	ARC5
Starting structure	2-automatic coloring	3-automatic coloring	5-automatic coloring
Main bridge	Embed a suitable 3-kernel family into the finite 2-kernel	Direct binary-decimation closure gives $K_2(a) \subseteq K_3(a)$	Collatz local density + dyadic thinness + base-5 pumping
Arithmetic input	Primitive-root behavior of 2 modulo powers of 3	Coefficient 3 fits the 3-kernel hierarchy	Oddness of 5^L and exact powers-of-two displacement of pumped values
Global compactness	Cobham $\Rightarrow$ eventual periodicity	Cobham $\Rightarrow$ eventual periodicity	Finite change set $\Rightarrow$ eventual constancy
Final propagation	Collatz identities collapse the period / positive domain	Same residual identities collapse the period	Doubling invariance transports tail value to every positive input
Cobham theorem	Used	Used	Not used in the main proof

The comparison suggests that “ARC in base b ” should not be expected to admit one uniform elementary proof. Bases that interact directly with the coefficients 2 and 3 may permit kernel transfer. Other odd bases may instead be approachable through local synchronization plus 2-adic pumping. A natural next program is to determine which parts of Theorem 5.1 hold for arbitrary odd bases b , and which parts of the Collatz local-density mechanism are base-independent.

11 Formal verification in Lean 4

The proof chain was developed in Lean 4 / Mathlib. The final entry point is

`arc5_final`,

whose hypotheses are exactly shortcut invariance and `ARCAutomaticByKernel 5 a`, and whose conclusion is existence of a single value on every positive natural input.

The human-readable proof above compresses a comparatively large formal development. The principal groups of files are as follows.

Formal component	Main verified content
<code>ARC5LocalDensityStage1-3E</code>	Parity-word construction; affine closed forms; universal translation-word coalescence; admissibility on dyadic cylinders; deterministic shortcut-iterate interface.
<code>ARC5LocalDensityStage4A-4Q</code>	Balanced paired refinement inside prescribed cylinders; finite simultaneous coalescence; synchronized local-density blocks for invariant colorings.
<code>ARC5LocalDensityStage4R-4T</code>	Natural-domain bridge; arbitrarily long monochromatic natural blocks; base-5 aligned blocks; finite-kernel/local-density handoff.
<code>ARC5Stage5A-5F</code>	Exact level kernels, digit-transition graph, strong dyadic synchronization, and cross-base dyadic agreement of same-level states.
<code>ARC5Stage5G-5I</code>	Disagreement sets, dyadic nowhere-thickness, automatic disagreement indicators, and the isolated topological pumping interface.
<code>ARC5Stage5J-5Q</code>	Geometric pumped families; MSD word pumping; positivity of the numerical increment; exact $2^s \times \text{odd}$ factorization; dyadic residue permutation and child filling; contradiction with nowhere-thickness.
<code>ARC5Stage5R-5S</code>	Infinite support gives arbitrarily long canonical base-5 support words; MSD topological pumping theorem; finite-kernel-to-MSD integration; unconditional finite support for nowhere-thick automatic sets.
<code>ARC5Stage5T-5U</code>	Automaticity of the adjacent-change indicator; strong dyadic synchronization makes the change set nowhere thick; therefore the change set is finite.
<code>ARC5Stage5V-5W</code>	Finite change set implies eventual constancy; powers-of-two invariance propagates the tail value to every positive integer.
<code>ARC5Final.lean</code>	Final packaged theorem <code>arc5_final</code> and pairwise positive-domain equality.

Cobham and external assumptions

Unlike ARC2 and ARC3, the ARC5 main chain does not invoke the Cobham principle. The only automaticity conversion needed in the topological pumping stage is the formally proved bridge from finite kernel to an MSD-DFAO presentation. Thus the central compactness step is internal: finite-state pumping plus the dyadic arithmetic of base-5 word values.

Axiom auditing

The final ARC5 files contain explicit `#print axioms` commands for the principal handoff theorems, including the local-density interfaces, pumping theorem, finite change set, eventual constancy, positive rigidity, and `arc5_final`. The retained build output for the shared automaticity bridge reports the standard dependencies `propext`, `Classical.choice`, and `Quot.sound`; no Cobham axiom is required by the ARC5 main route. The source-level development used for this manuscript contains completed theorem proofs rather than an intentionally postulated ARC5 rigidity principle.

Representative theorem names

Representative Lean theorem names matching the paper’s chain include

- `arcFiniteNaturalInvariantBlock_pow5`,
- `arcFiniteNaturalAlignedInvariantBlock_pow5`,
- `arc5_stage5F_cross_base_collision`,
- `arc5_level_disagreement_nowhereThick`,
- `arc5_stage5R_msd_topological_pumping`,
- `arc5_stage5S_topological_pumping_principle`,
- `arc5_changeSet_finite`,
- `arc5_eventually_constant`,
- `arc5_stage5W_positive_rigidity`,
- `arc5_final`.

Disclosure of generative-AI assistance

Generative AI (OpenAI ChatGPT, GPT-5.6 Sol) was used in preparing this manuscript for literature-search assistance, theorem formulation, adversarial checking of proof steps, editorial drafting, assistance with Lean 4 proof-code development, diagnosis of compilation errors, and post-formalization auditing. The AI system is not an author and is not used as a bibliographic source. Responsibility for the mathematical claims, references, formalization, and final manuscript rests with the human author.

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